

SpaCy

- Overview of SpaCy and Its Features
 - Advantages of using SpaCy
 - SpaCy vs NLTK and other NLP Libraries
 - Installing SpaCy (`pip install spacy`)
 - Loading and Exploring SpaCy Models
 - Pre-trained models: `en_core_web_sm`, `en_core_web_md`, `en_core_web_lg`
 - Exploring model properties
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2. Text Preprocessing with SpaCy

2.1 Tokenization

- Tokenization in SpaCy (`nlp.tokenizer`)
- Word and Sentence Tokenization
- Token Attributes in SpaCy: `text`, `lemma_`, `pos_`, `dep_`

2.2 Text Cleaning

- Removing Punctuation and Special Characters
- Handling Special Tokens: Stopwords, URLs, Numbers
- Case Normalization (Lowercasing)
- Custom Tokenization Rules

2.3 Lemmatization and Stemming

- Lemmatization in SpaCy (e.g., `token.lemma_`)
 - Difference between Lemmatization and Stemming
 - Handling irregular forms (e.g., `run` -> `ran`)
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3. Part-of-Speech (POS) Tagging and Named Entity Recognition (NER)

3.1 Part-of-Speech (POS) Tagging

- POS Tagging and its Applications
- SpaCy's POS Tagging (`token.pos_`, `token.tag_`)
- POS Tagging with Pre-trained Models
- Understanding Universal Dependencies

3.2 Named Entity Recognition (NER)

- Introduction to Named Entity Recognition
 - Pre-trained SpaCy Model for NER
 - Entity Types (e.g., PERSON, ORG, LOC)
 - Custom NER Models and Training NER with SpaCy
 - Working with `ents` and extracting entities from documents
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4. Dependency Parsing and Syntactic Analysis

4.1 Introduction to Dependency Parsing

- What is Dependency Parsing?
- Understanding Syntactic Relations: Subjects, Objects, etc.
- Visualizing Parse Trees in SpaCy (`displacy.render()`)
- SpaCy's Dependency Parser (`token.dep_`)

4.2 Working with Syntactic Trees

- Understanding Tree Structures and Dependency Arcs
- Root, Children, and Parent Tokens in a Sentence
- Visualizing Dependency Trees using SpaCy's built-in tools

4.3 Sentence Segmentation

- Sentence Detection and Segmentation in SpaCy
 - Custom Sentence Segmentation
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5. Word Vectors and Word Similarity

5.1 Word Embeddings in SpaCy

- Introduction to Word Embeddings
- Using Pre-trained Word Vectors with SpaCy

- Loading word vectors with `spacy.load()`
 - Word2Vec and GloVe embeddings
- Performing Word Similarity Calculations (`token.similarity()`)
- Analyzing and Visualizing Word Vectors

5.2 Visualizing Word Vectors

- Using t-SNE to Reduce Dimensions
 - Plotting Word Vectors with Matplotlib and Plotly
 - Visualizing Word Relationships with SpaCy's `displacy`
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6. Text Classification and Feature Extraction

6.1 Text Classification with SpaCy

- Introduction to Text Classification
- Text Classification Pipelines in SpaCy
- Custom Classifier with SpaCy
 - Creating a Text Classification Model with `TextCategorizer`
- Handling Imbalanced Datasets and Evaluating Models

6.2 Feature Extraction

- Extracting Features for Classification Tasks
 - Using SpaCy for Feature Extraction (BoW, TF-IDF)
 - Implementing a Custom Feature Extraction Pipeline
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7. Custom NLP Pipelines and Components

7.1 SpaCy Pipelines

- What are SpaCy Pipelines?
- Building and Customizing NLP Pipelines
 - Adding custom components to the pipeline
 - Working with pre-existing components like NER, POS Tagger, etc.

7.2 Custom Components and Token Analyzers

- Writing Custom Components in the Pipeline

- Creating Token Analyzers (e.g., Text Classification or Sentiment Analysis)
- Working with `@Language.component` decorator

7.3 Custom Tokenization

- Implementing Custom Tokenization Rules
 - Handling Edge Cases in Tokenization (URLs, hashtags, special characters)
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8. Named Entity Recognition (NER) with SpaCy

8.1 Introduction to NER

- Understanding NER and Its Use Cases
- Using SpaCy's Pretrained NER Models
- Extracting Named Entities from Text
- Entity Recognition in Different Languages

8.2 Training a Custom NER Model

- Preparing Data for NER Training
 - Annotating Data (using `Prodi.py` or other tools)
 - Training a Custom NER Model with SpaCy
 - Fine-tuning the NER Model on Custom Data
 - Evaluating the Performance of NER Models
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9. Advanced Topics in SpaCy

9.1 Word and Sentence Vectors

- Working with `doc.vector` for Sentence Embeddings
- Using Pre-trained Sentence Models
- Sentence Similarity using SpaCy's Doc Vectors

9.2 SpaCy + Machine Learning

- Using SpaCy for Preprocessing in ML Pipelines
- Combining SpaCy with Scikit-learn for NLP Tasks
- Integrating SpaCy's Feature Extraction with SVM, Naive Bayes, etc.

9.3 Transfer Learning and Fine-tuning

- Fine-tuning SpaCy's Pretrained Models (e.g., BERT, RoBERTa)
 - Transfer Learning for Text Classification, NER, and Text Generation
 - Using Hugging Face with SpaCy for Advanced Models
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10. Text Generation and Language Models

10.1 Text Generation with SpaCy

- Introduction to Text Generation
- Using SpaCy for Simple Text Generation
- Building a Markov Chain Text Generator

10.2 Language Models with SpaCy

- Introduction to Language Models and Their Use
 - Fine-tuning Pretrained Language Models (e.g., GPT-2) for Text Generation
 - Implementing Neural Language Models with SpaCy
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11. SpaCy and Deep Learning

11.1 Introduction to Deep Learning with SpaCy

- Understanding Deep Learning Architectures (RNN, LSTM, CNN)
- Using SpaCy with Deep Learning Libraries (TensorFlow, PyTorch)

11.2 Training a Neural Network Model for NLP

- Using SpaCy with TensorFlow/PyTorch for Text Classification
 - Implementing LSTM Networks for NLP Tasks using SpaCy
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12. Evaluation Metrics in SpaCy

- Accuracy, Precision, Recall, and F1 Score for NLP Models
- Evaluating Named Entity Recognition and Text Classification Models
- Using SpaCy's `scorer` for Model Evaluation

- Cross-validation in SpaCy
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13. Real-World NLP Projects with SpaCy

13.1 Beginner Projects

- Sentiment Analysis using SpaCy
- Named Entity Recognition on Texts (News Articles, Tweets)
- Build a POS Tagging Pipeline for Custom Text

13.2 Intermediate Projects

- Build a Text Classification Model for Email Filtering
- Implement a Question Answering System with SpaCy
- Create a Custom Named Entity Recognition Model for Legal Documents

13.3 Advanced Projects

- Implement a Text Summarization Pipeline with SpaCy
 - Build a Chatbot or Conversational Agent using SpaCy
 - Train a Multi-lingual NER System with SpaCy
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14. Deployment and Optimization

14.1 Model Serialization and Saving

- Saving and Loading Models in SpaCy
- Exporting and Importing Models with `spacy.load()` and `spacy.blank()`

14.2 Optimizing SpaCy Pipelines

- Optimizing Pipeline Components for Speed and Efficiency
- Using `spacy.displacy` for Visualizing Models in Production
- Integrating SpaCy into Web and Enterprise Applications

